

King Faisal University
College of Computer Sciences and Information Technology
Computer Science Department

[CS-210]: [Programming Methodologies]
Section: []

Lab [04]: [Nested if else and Switch Statements]

Date: []

Answer Sheet

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Computer Number:

Part 1 — Guided Debugging & Code Analysis

Error Identification & Correction Table

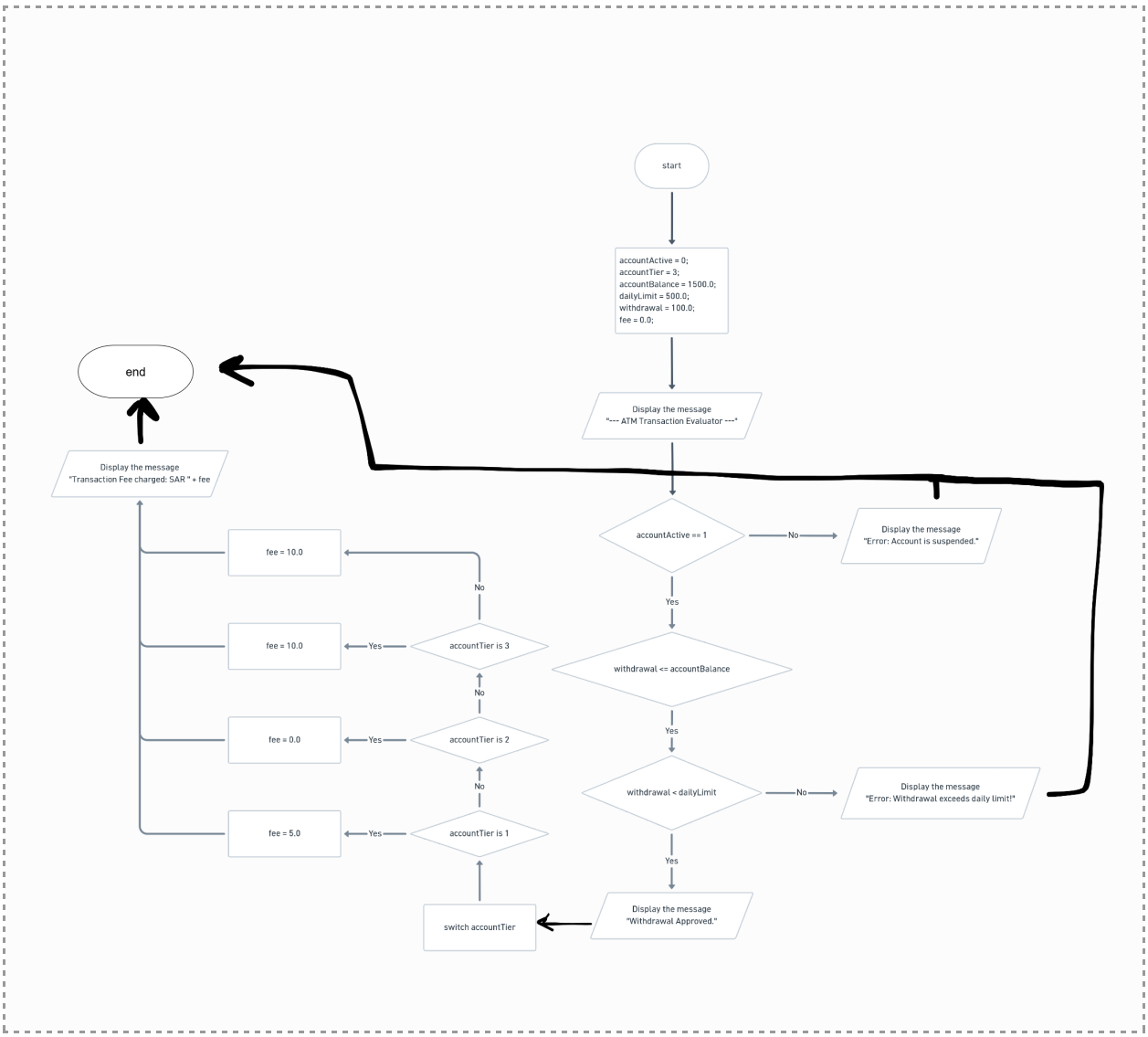
Fill in the table below by identifying the errors present in the code. One row is dedicated to each error.

Line #	Code Snippet with Error	Error Description (Student Fill)	Error Type (Syntax/Logical)	Corrected Code (Student Fill)
7	# 1: Regular, 2: Premium, 3: VIP	# is used to implement a comment instead of //	SyntaxError	// 1: Regular, 2: Premium, 3: VIP
22	switch (accountBalance) {	It should be accountTier instead of accountBalance	SyntaxError	switch (accountTier) {
18	if (withdrawal > dailyLimit) {	it should be approved only if the withdrawal LESS than dailyLimit, NOT BIGGER	LogicalError	if (withdrawal < dailyLimit) {
24	fee = 5.0;	There are not any breaks. Thus, it will re-assign fee until a break is founded	LogicalError	fee = 5.0; break;

16	if (accountActive == 1)	It doesn't have any braces; leading to not execute properly	LogicalError	if (accountActive == 1)
				{
				}

Flowchart Diagram

Draw the flowchart of the given problem in the box below.



Corrected Source Code

Paste your complete, working, and corrected C++ source code in the box below.

```
// ATM Withdrawal & Fee Evaluator (Flawed Code)
#include <iostream>
using namespace std;

int main() {
    int accountActive = 0;   // 1 for Active, 0 for Suspended
    int accountTier = 3;     // 1: Regular, 2: Premium, 3: VIP
    double accountBalance = 1500.0;
    double dailyLimit = 500.0;
    double withdrawal = 100.0;
    double fee = 0.0;

    cout << "--- ATM Transaction Evaluator ---" << endl;

    // Check account status and withdrawal request
    if (accountActive == 1)
    {
        if (withdrawal <= accountBalance) {
            if (withdrawal < dailyLimit) {
                cout << "Withdrawal Approved." << endl;

                // Determine transaction fee based on account tier
                switch (accountTier) {
                    case 1:
                        fee = 5.0;
                        break;
                    case 2:
                        fee = 2.0;
                        break;
                    case 3:
                        fee = 0.0;
                        break;
                    default:
                        fee = 10.0;
                }
                cout << "Transaction Fee charged: SAR " << fee << endl;
            }
            else {
                cout << "Error: Withdrawal exceeds daily limit!" << endl;
            }
        }
    }
    else {
        cout << "Error: Account is suspended." << endl;
    }

    return 0;
}
```

Execution Screenshot

Paste a screenshot of your working program output here showing a successful temperature conversion.

Active: 1, Tier: 1
Balance: 1500, Limit: 500
Withdrawal: 300

```
Output:

--- ATM Transaction Evaluator ---
Withdrawal Approved.
Transaction Fee charged: SAR 5

Active: 1, Tier: 2
Balance: 1500, Limit: 500
Withdrawal: 700

Output:

--- ATM Transaction Evaluator ---
Error: Withdrawal exceeds daily limit!

Active: 0, Tier: 3
Balance: 1500, Limit: 500
Withdrawal: 100

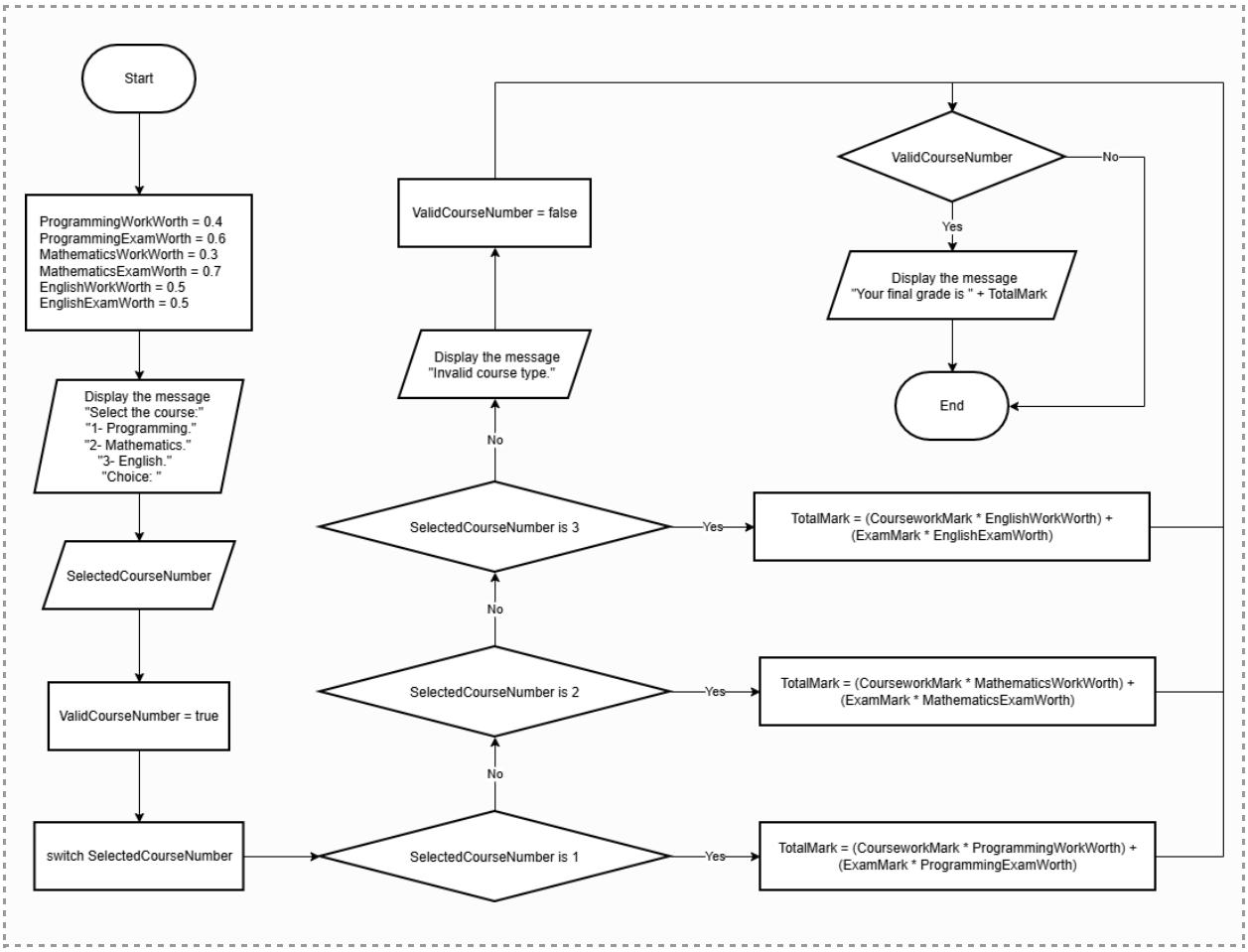
Output:

--- ATM Transaction Evaluator ---
Error: Account is suspended.
```

Part 2 — Independent Program Implementation

Flowchart Diagram

Draw the flowchart of the given problem in the box below.



C++ Source Code:

Paste your complete C++ source code for the VAT calculator in the box below.

```cpp
#include <iostream>
using namespace std;

int main() {

    int SelectedCourseNumber;
    const float ProgrammingWorkWorth = 0.4, ProgrammingExamWorth = 0.6;
    const float MathematicsWorkWorth = 0.3, MathematicsExamWorth = 0.7;
    const float EnglishWorkWorth = 0.5, EnglishExamWorth = 0.5;
    float CourseworkMark, ExamMark;
    float TotalMark;

    /*
    I've changed the order of user input insertion
    to organize the code. Instead of listing the
    courses first then entering marks, i did the
    opposite. I will left the code that is ordered
    like in the homework after the main() function.
    */

    cout << "Enter the coursework mark: ";
    cin >> CourseworkMark;

    cout << "Enter the exam mark: ";
    cin >> ExamMark;

    cout << endl;
    cout << "Select the course:\n\n";

    cout << "1- Programming.\n";
    cout << "2- Mathematics.\n";
    cout << "3- English.\n";

    cout << "\nChoice: ";
    cin >> SelectedCourseNumber;

    bool ValidCourseNumber = true;

    switch (SelectedCourseNumber)
    {
    case 1: // Programming is chosen.
        TotalMark = (CourseworkMark * ProgrammingWorkWorth) + (ExamMark * ProgrammingExamWorth);
        break;
    case 2: // Mathematics is chosen.
        TotalMark = (CourseworkMark * MathematicsWorkWorth) + (ExamMark * MathematicsExamWorth);
        break;
    case 3: // English is chosen.
        TotalMark = (CourseworkMark * EnglishWorkWorth) + (ExamMark * EnglishExamWorth);
        break;
    default: // Wrong course number.
        cout << "Invalid course type.\n";
        ValidCourseNumber = false;
    }

    if (ValidCourseNumber)
        cout << "Your final grade is " << TotalMark << endl;

    return 0;
}

/*
* if you insist, Doctor, this is the code
exactly as ordered in the homework. It will
start from listing the courses to be brief:

    ...
    cout << "Select the course:\n\n";

    cout << "1- Programming.\n";
    cout << "2- Mathematics.\n";
    cout << "3- English.\n";

    cout << "\nChoice: ";
    cin >> SelectedCourseNumber;

    bool ValidCourseNumber = true;
```

```
    switch (SelectedCourseNumber)
    {
    case 1: // Programming is chosen.
        cout << "Enter the coursework mark: ";
        cin >> CourseworkMark;

        cout << "Enter the exam mark: ";
        cin >> ExamMark;

        TotalMark = (CourseworkMark * ProgrammingWorkWorth) + (ExamMark * ProgrammingExamWorth);
        break;
    case 2: // Mathematics is chosen.
        cout << "Enter the coursework mark: ";
        cin >> CourseworkMark;

        cout << "Enter the exam mark: ";
        cin >> ExamMark;

        TotalMark = (CourseworkMark * MathematicsWorkWorth) + (ExamMark * MathematicsExamWorth);
        break;
    case 3: // English is chosen.
        cout << "Enter the coursework mark: ";
        cin >> CourseworkMark;

        cout << "Enter the exam mark: ";
        cin >> ExamMark;

        TotalMark = (CourseworkMark * EnglishWorkWorth) + (ExamMark * EnglishExamWorth);
        break;
    default: // Wrong course number.
        cout << "Invalid course type.\n";
        ValidCourseNumber = false;
    }

    if (ValidCourseNumber)
        cout << "Your final grade is " << TotalMark << endl;

    return 0;
*/
```

Execution Screenshot:

Paste a screenshot of your compiled VAT calculator running successfully with test data.

```
The following output is according to the “homework-order code”, which is
commented at the end of the code i’ve implemented.


Select the course:

1- Programming.
2- Mathematics.
3- English.

Choice: 1
Enter the coursework mark: 70
Enter the exam mark: 80
Your final grade is 76
```